



Apple announcements ranked

All announcements at WWDC in order of importance, starting from the transition to in-house silicon. <https://digit.in/jul20-31>



Quantum language

A language for quantum computers that is as intuitive as classical computer languages. <https://digit.in/jul20-32>

could be something you wouldn't have thought in your wildest dream. It can be a router's firmware, a CCTV camera, an iPod or your old point-and-shoot Sony Cybershot and the list is neverending. Oh, and the evidence might not be physical at all! Image forensics, audio forensics, network forensics, etc. are all sub-branches of DF, so beware because these pictures (and ahem, maybe the network traffic also) are worth more than a thousand words and beyond.

Analysis

The big guns come out here. After the acquisition is done, imaging of the evidence is performed. That means the firmware retrieved from any source is first converted into a format that can be analysed with tools. Imaging is necessary to create perfect replicas of the evidence, without losing any piece of information. This is done by creating a sector (the smallest unit of a hard drive) level duplicate of the media, also known as a "forensic duplicate". A special device called a "write-block" device is used to prevent any modification during the imaging process. After the imaging is done, the original and the duplicate data is hashed (using algorithms like SHA-1 or MD5) and compared to check the accuracy. The next step is the process of swimming through tons of information to find the bad actors, hidden somewhere. Doing this manually is a task that is near impossible to achieve and hence a wide array of tools are used. We have divided the tools into certain categories (we will cover only major tools), to make their roles clearer. The tools are-

Computer Forensics

This is the branch that focuses on computers and digital media evidence and involves the analysis of the disk images, we talked about above. Here's the list of the very best tools in the industry:

- The Sleuth Kit: Also known as TSK, The Sleuth Kit is a curated

collection of tools that allows forensic analysis of both Microsoft and UNIX file systems and disks. An open source initiative, TSK was majorly written and is maintained by Brian Carrier, who is a digital investigator and has garnered around 1.2k stars on GitHub [<https://digit.in/sleuthkit>]. A large number of file system formats are supported by TSK, including but not limited to NTFS, FAT/ExFAT, UFS 1/2, Ext2 etc. TSK is written mostly in C and Perl, with some code in Python too. And for those of you who are a bit wary of the command line, all hope is not lost since there's actually a GUI front available for TSK, known as Autopsy and interestingly, it also support plugins that can be used to extend TSK's ability even further. Any old timers of the forensics industry might remember The Coroner's Kit and TSK is the official successor of the platform.

- Volatility Framework: Memory analysis doesn't get sharper than this. Based on the foundation of years of published research on advanced memory analysis, Volatility was released at Blackhat 2014 and changed the game forever. Volatility introduced and showed people how powerful it is to analyse the runtime state of a system, in contrast to static analysis, by sifting through the data found in the volatile storage/memory, that is, the RAM. On top of that, Volatility is cross-platform, able to analyse Windows, MacOS and Linux-based computers on a memory level. Written in Python, this too is open source and available on GitHub. You can straightaway download and start using it if you feel like! Volatility requires a RAM dump that needs to be analysed and a profile associated with that dump. Volatility can then perform numerous analysis and spit out the service, processes, etc running at the

time of the incident (sometimes this results in uncovering cached passwords hashes in the memory dump of the operating system, that can later be cracked). Again, a GUI is available, but unlike The Sleuth Kit's independent desktop application, Volatility has a web based GUI, called eVOLve. It is an indispensable tool, on which military, law enforcement, academia and commercial investigators rely heavily upon.

- Foremost/Scalpel: Both these are tools known as "file carvers". File carving is the process of looking through huge data streams (data streams can be images, compressed archives, etc) and extracting out files solely based on the file formats and no other meta-data. This is possible to achieve in many different ways, but the most popular way is to look for "magic numbers". Each file type has distinct hex values at the starting of their binary, which are called file signatures or "magic numbers" and they are constant for any file of that extension. For example, if you open up a ".png" file in a hex editor and check the first values, you will always find them to be "89 50 4E 47" and if you don't, then it's not a png file. These types of challenges are quite frequent in Capture The Flag (CTF) competition where a file with garbled up magic numbers contains a flag. So the next time you tackle a problem like this, we hope you keep this article in mind! One more popular tool falling in the same ballpark is binwalk.

Network Forensics:

This category deals with the network traffic and analysis of the packets that get transferred during any sort of network activity, spanning over protocols like HTTP, HTTPS, FTP, etc (sometimes even printer and USB network streams!)

- Wireshark: We can't have a discussion about network forensics